

ngspice Linear Solvers — KLU vs. Sparse 1.3

Behavior, defaults, and limitations across DC / AC / transient / noise

Ngspice + OpenVAF Enhancements project

July 2026

Contents

ngspice linear solvers — KLU vs. Sparse 1.3 (behavior, defaults, limitations)	1
TL;DR	1
Selecting and confirming the solver	2
Noise and pole-zero under KLU (Enhancement-113)	2
KLU discrepancies — all resolved	3
The dual-solver test harness	3

ngspice linear solvers — KLU vs. Sparse 1.3 (behavior, defaults, limitations)

This build of ngspice-46 ships two direct linear solvers, and they are not fully interchangeable across analysis types. This note records exactly how they differ — which is the default, how to select and confirm each, and where KLU falls short — based on a direct read of the source and a solver-by-solver sweep of the whole [examples/](#) suite.

TL;DR

	KLU	Sparse 1.3
Availability in this build	Compiled in (SuiteSparse, statically linked — 44 <code>klu_*</code> symbols in the binary)	Always present (ngspice's own solver)
Default?	No	Yes — this build defaults to Sparse 1.3
How to select	<code>.option klu</code>	<code>.option sparse</code> (or just the default)
DC op / DC sweep	✓ correct	✓ correct
AC	✓ correct	✓ correct
Transient	✓ correct (one caveat below)	✓ correct
Noise (.noise)	✓ correct (since E-113)	✓ correct
Pole-zero (.pz)	✓ correct (single-ended E-113; balanced-output + complex-root determinant E-171/172)	✓ correct
Sensitivity (.sens , DC & AC)	✓ correct (since E-114)	✓ correct
Distortion (.disto)	✓ correct (since E-115)	✓ correct
Periodic steady state (.pss)	✓ correct (since E-118)	✓ correct
Periodic small-signal (.pac/.pnoise/.pxf/.psp)	✓ correct	✓ correct
Transient checkpoint/restart (savestate/loadstate)	⚠ Sparse-only (E-131)	✓ correct

The periodic small-signal analyses (PAC / Pnoise / PXF and the E-132 periodic S-parameters `.psp`) build a dense $(2M+1)N$ harmonic conversion matrix and solve it with a standalone dense complex LU (`pss_csolve`) that is independent of the sparse linear solver — so they inherit the solver only through the underlying PSS, which runs correctly under both since E-118 (KLU re-factors every shooting step, so PSS is *slower* under KLU but not wrong). The E-133 two-tone qpss and the E-134 frequency-domain hb (harmonic balance) are likewise solver-independent

— qpss drives a transient and direct-DFTs it; hb does a dense complex Newton on the conversion matrix and uses the sparse solver only to *read* the periodic $G(t)/C(t)$ off the device matrix. That read is the one solver-specific detail: Sparse uses `spSetComplex`, KLU needs the complex-CSC binding (`DEVbindCSCComplex`), exactly as the PAC harmonic extraction does — hb carries the same `#ifdef KLU` branch, so it is verified bit-identical under `.option klu` and `.option sparse` (a bare hb also copies the task's KLU mode before building the matrix, so `.option klu` takes effect without a prior analysis). Transient checkpoint/restart ([Enhancement-131](#)) is the one feature that is genuinely Sparse-only: on the restore path KLU's symbolic/numeric factorization objects are absent (they are only built during a full run's operating point), so `savestate/loadstate` reject `.option klu` with a clear message rather than crash.

Practical guidance: leave the default (Sparse 1.3) unless you have a specific reason to switch. Sparse 1.3 runs every analysis in the suite. Since [Enhancement-113](#) KLU also runs noise and single-ended pole-zero correctly, and since [Enhancement-114](#) it runs DC/AC sensitivity, and since [Enhancement-115](#) distortion (`.disto`) correctly; since [Enhancement-171/172](#) the pole-zero path is fully KLU-correct too (complex-plane determinant, balanced output, full-partial-pivot fallback) — no analysis is Sparse-only under KLU any more (the transient checkpoint/restart *feature* of E-131 still is, per the table above). Reach for `.option klu` on large, sparse DC/AC problems where KLU's ordering and factorization are faster, and — because its symbolic ordering is computed once and cannot re-pivot dynamically like Sparse — expect it to be less forgiving of a near-singular Jacobian on a stiff transient edge, which the dissipative Gear integrator avoids (see the [former opamp741 discrepancy](#) below).

Tuning KLU ([Enhancement-152](#)): KLU's reordering and scaling, previously hard-coded, are now `.options` — `klu_ordering=amd|colamd` (fill-reducing ordering), `klu_scale=none|sum|max` (row equilibration), and `klu_btf=on|off` (block-triangular-form permutation), with defaults `amd/max/on`. They change only *how* the matrix factors, never the solution (verified physically identical to $\sim 1e-14$ across all settings), and matter only on unusual matrix structures (a different ordering can reduce fill) or badly-scaled matrices. The `klu_memgrow_factor` knob was also fixed (it had silently collapsed to a boolean).

Selecting and confirming the solver

The netlist option is a simple flag:

```
.option klu      * use KLU (SuiteSparse)
.option sparse   * use the legacy Sparse 1.3 (this is also the default)
```

Because unknown `.option` keywords are silently ignored by ngspice, "it ran without complaint" is *not* evidence the solver changed. To confirm which solver is actually active, use the interactive option command, which prints the live `CKTklumODE`:

```
.control
run
option      * prints, among other things: "Matrix solver: KLU" or "Sparse 1.3"
.endc
```

On the committed bin/ binary this reports **Sparse 1.3** for a bare deck and for `.option sparse`, and **KLU** only when `.option klu` is given — confirming Sparse 1.3 is the default here.

Noise and pole-zero under KLU ([Enhancement-113](#))

ngspice used to refuse both outright (`noisean.c / pzan.c` printed "not (yet) supported with 'option KLU'"). The real reason for noise was subtler than "not wired up": noise uses the adjoint method — it solves the *transposed* system $A^T \cdot x = e$ via `SMPcaSolve` — and `SMPcaSolve`'s KLU branch was calling the non-transposed `klu_z_solve` where its Sparse branch calls `spSolveTransposed`. That silently produced the wrong noise for any asymmetric matrix (every circuit with a transistor or controlled source), which is why it was disabled rather than merely slow.

[Enhancement-113](#) fixes the adjoint solve (`klu_z_tsolve`) and lifts the guards, so under KLU:

- Noise is correct — verified identical to Sparse on resistor thermal noise, asymmetric VCCS circuits, OSDI device models, and integrated totals. (`.sp` S-parameters share the same adjoint solve and are corrected too.)
- Single-ended pole-zero (grounded output reference) runs correctly.
- Balanced-output pole-zero (non-grounded reference) was Sparse-only at first: its zeros phase folds columns at solve time, which Sparse survives via dynamic Markowitz re-ordering but KLU's fixed symbolic factorization could not. [Enhancement-172](#) closed it — `CKTpzSetup` reserves the union pattern before `COO→CSC` conversion and `SMPcAddCol` gained a merge-walk KLU branch — and replaced the out-of-range pivot-tolerance fallback with full partial pivoting (`tol=1.0`), which also cured spurious far-field roots and a twin-T conjugate-pair stall. [Enhancement-171](#) had first fixed the KLU complex determinant itself (mixed real/complex pivot products and permutation parity — silent garbage for complex roots).

[Enhancement-114](#) then fixes sensitivity under KLU. Sensitivity builds an auxiliary perturbation matrix `delta_Y` that is a plain Sparse matrix (it is only multiplied, never factored); two KLU setup blocks were gated on the *main* matrix's flag and dereferenced the NULL `delta_Y->SMPkluMatrix`, segfaulting on every DC/AC .sens deck. Gating them on `delta_Y`'s own flag keeps it Sparse under KLU too, so both DC and AC sensitivity now match Sparse exactly.

- Distortion (`.disto`) was Sparse-only — `distoan.c` had no KLU code, so under `.option klu` the (complex) distortion solves ran against a matrix left in real mode and silently produced no output. Surfaced while validating E-114. [Enhancement-115](#) fixes it: it converts the KLU matrix `real↔complex` around the distortion solve loop (exactly as `acan.c` does for AC), so `.disto` now matches Sparse bit-for-bit.

KLU discrepancies — all resolved

Beyond balanced-output pole-zero (a genuine "not (yet) supported" *skip*, not a wrong result), no example now produces a different result under KLU than under Sparse. `KLU_XFAIL` is empty; the harness runs every example under both solvers and expects agreement.

`opamp741` — was the last one; the real cause was the integration method, not KLU. The transistor-level [opamp741 example](#) (a μ A741 from ~70 lines of Verilog-A BJT) used to abort under KLU on its large-signal slew test (`pulse(-5 5 ...)` into the follower): output-stage transistors switch off at the slewing edge, their transconductances collapse, KLU declares the Jacobian singular (`x1.o1/o2/b34/cm`), the timestep collapses, and ngspice aborted at $t \approx 2.03 \mu\text{s}$ — while Sparse ran the full `tran 20n 80u` cleanly.

The near-singular Jacobian at the edge was manufactured by the default trapezoidal integrator, which is non-dissipative and *rings* on the stiff feedback slew, driving the output transistors hard off. It is that ringing — not KLU's linear solve — that produced the collapse: Sparse's dynamic Markowitz re-ordering happened to survive the near-singular step where KLU's fixed symbolic ordering could not, but the step should never have been that singular. Switching the two transient decks to Gear (`.option method=gear`, dissipative BDF-2) damps the ringing so the edge is never near-singular: the slew now runs to completion under both solvers and the results agree to ~8 sig figs (final `V(out)` — 3.3153833712 KLU vs — 3.3153833649 Sparse), with every figure of merit (`Aol`, `fu`, `PM`, slew rate, offset, swing) identical between the solvers. `opamp741` was removed from `KLU_XFAIL` accordingly.

(This corrects the earlier reading of this case as an unfixable KLU linear-solver limitation needing a hybrid Sparse-fallback: the earlier pivoting/ordering knobs [E-116] left the *trapezoidal* run's abort unchanged because they addressed the symptom, not the ringing that caused it. The genuinely structural KLU issues were the two below, which E-116 did fix.)

Two former discrepancies, now fixed (E-116). `groundcontrib` (node-to-ground `V(p,gnd) <+ 1.5 read v(p)=0` under KLU instead of 1.5) and `hierbranch` (hierarchical branch-*current* probes read 0) were both structural, not numerical: an OSDI internal node that appears in no Jacobian entry — a ground reference whose branch contribution drops its column — was allocated its own all-zero solver row, which Sparse tolerates but KLU treats as structurally singular. [Enhancement-116](#) ties such decoupled internal nodes to ground at setup, so both now match Sparse under KLU.

The dual-solver test harness

Every `verify_*.py` in `examples/` is run under both solvers automatically. Each script calls `check_both_solvers(__file__)` (right after importing `_setup`); on a normal run that re-executes the script once per solver — injecting `.option sparse / .option klu` into every ngspice deck — and prints a combined verdict:

```
=== BOTH-SOLVER RESULT [ceil]: sparse=PASS klu=PASS => OK ===
```

KLU's one remaining unsupported analysis (balanced-output pole-zero) is auto-detected and reported SKIP; `KLU_XFAIL` is now empty (`opamp741`, its last member, was resolved — see above). A separate `SPARSE_ONLY` registry (`{rfanalyses, rfpss}`) marks the heavy periodic-steady-state examples whose KLU pass is not *wrong*, just slow — KLU re-factors every PSS shooting step, so a 1024-sample `.pss` that is a couple of minutes under Sparse becomes 10–15 min under KLU. Those run Sparse-only by default (reported `klu=SKIP`); their KLU correctness is a property of the ngspice source, verified once when [E-118](#) landed, not something worth re-paying every sweep. Env escape hatches: `NGSPICE_SOLVER=klu|sparse` runs once under one solver; `NG_BOTH=0` disables the dual run; `NG_SLOW_KLU=1` forces the skipped heavy-PSS KLU pass back on (to re-check the E-118 KLU-PSS fix on demand).

Sweep result (all 101 machine-checkable scripts): 101/101 OK — every example is `sparse=PASS` with `klu` in `{PASS, SKIP, XFAIL}`, i.e. the two solvers agree wherever KLU is applicable. (The `linesearch` example initially *crashed* under KLU, which [Enhancement-112](#) fixed; [Enhancement-113](#) then moved 10 examples from KLU-skipped to KLU-passing by enabling noise and single-ended pole-zero; [Enhancement-114](#) fixed the AC/DC-sensitivity crash under KLU, and — by fixing the deck-injector restore bug — un-masked two more XFAILs (`groundcontrib`, `hierbranch`);

[Enhancement-115](#) then fixed distortion, so the analyses example now runs fully under KLU; and [Enhancement-116](#) fixed groundcontrib and hierbranch at their shared root cause; and opamp741 — the last XFAIL — was resolved by running its stiff transient under Gear rather than the default trapezoidal method, so **KLU_XFAIL** is now empty.)

Conclusion: for DC, AC, transient, noise, single-ended pole-zero, sensitivity, and distortion the two solvers agree across the entire suite (KLU_XFAIL is empty), while balanced-output pole-zero is Sparse-1.3-only by ngspice design. Sparse 1.3, the default, covers everything — which is why it is the default and why the dual-solver harness keys correctness off it.